# Electrochemical equilibrium

Week 6 · Onsite · 180 minutes

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# Q and K for the same reaction

For $\mathrm{A_2(g)} \rightleftharpoons 2\mathrm{A(g)}$, use the dimensionless activity ratio $a_{\mathrm{A}}^2 / a_{\mathrm{A_2}}$.

**Reaction quotient, $Q$.** Calculate the ratio using the activities in the mixture *now*.

**Equilibrium constant, $K$.** At equilibrium the ratio equals $K$. Its value is fixed at a given temperature for the written reaction and chosen standard states.

| If $Q < K$ | If $Q = K$ | If $Q > K$ |
|---|---|---|
| The forward reaction forms A. | The reaction is at equilibrium. | The reverse reaction forms $A_2$. |

# Does doubling the pressure always double Q?

**A question from Week 5.** How does the pressure argument change for a real gas?

Start with the ideal-gas reaction $\mathrm{A_2(g)} \rightleftharpoons 2\mathrm{A(g)}$. At 350 K, $K = 3.953$ for $p^\circ = 100$ kPa. Compress the mixture from 100 to 200 kPa at fixed temperature. Consider the instant before the composition responds.

| Starting value | Predict $Q$ immediately after compression | Which way will the reaction move? |
|---|---|---|
| $Q = K = 3.953$ | Calculate on paper | Toward A or A2? |
| $Q = 1.000$ | Calculate on paper | Toward A or A2? |

Take one minute. Compare your two answers with a neighbor.

# Compare the new quotient with K

At fixed mole fractions, the ideal-gas quotient for this reaction is proportional to pressure.

$$Q = \frac{y_{\mathrm{A}}^2}{y_{\mathrm{A_2}}} \frac{P}{p^\circ}$$

| Before compression | Immediately after | Response at 350 K |
|---|---|---|
| $Q = 3.953$ | $Q = 7.906 > K$ | More A2 forms. |
| $Q = 1.000$ | $Q = 2.000 < K$ | More A forms. |

As the reaction proceeds, the mole fractions change. At the new equilibrium, $Q = K = 3.953$ in both cases.

# A real gas uses fugacity in each activity

At a specified $T$, $P$, $\mathbf{y}$, a mixture equation of state gives each component's fugacity coefficient $\phi_i$. With an ideal-gas standard state,

$$a_i = \frac{f_i}{p^\circ} = \frac{\phi_i y_i P}{p^\circ}, \qquad Q = \frac{a_{\mathrm{A}}^2}{a_{\mathrm{A}_2}}.$$

For this dissociation, the multiplier on the ideal-gas estimate is $\phi_{\mathrm{A}}^2 / \phi_{\mathrm{A}_2}$. The coefficients can change during compression and as the reaction proceeds. The same $T$, written reaction and standard states give the same $K$.

**Try it.** At 350 K and 200 kPa, use $y_{\mathrm{A}} = 0.80$, $y_{\mathrm{A}_2} = 0.20$. For this illustrative state take $\phi_{\mathrm{A}} = 0.90$ and $\phi_{\mathrm{A}_2} = 0.95$. Calculate both activities and predict the direction.

# Work the real-gas example on paper

The pressure ratio is $200/100 = 2$. At the specified composition,

| Quantity | Calculation | Value |
|---|---|---|
| $a_{\mathrm{A}}$ | $0.90(0.80)(2)$ | 1.44 |
| $a_{\mathrm{A_2}}$ | $0.95(0.20)(2)$ | 0.38 |
| $Q$ | $1.44^2/0.38$ | 5.46 |

The ideal-gas estimate at this same state is $6.40$. Both exceed $K = 3.953$, so A2 forms. To find the final extent for the real gas, update the composition and both fugacity coefficients until $Q = K$. Lab 12 solves the ideal-gas case.

# The 180-minute class

| Class time | Activity |
|---|---|
| 0–15 min | Answer the Week 5 pressure question and check a real-gas calculation |
| 15–35 min | Revisit Weeks 2–5 in pairs with paper and a calculator |
| 35–60 min | Balance a cell reaction and calculate its potential by hand |
| 60–70 min | Demonstrate the concentration cell in Lab 13 |
| 70–80 min | Break |
| 80–150 min | Pair investigation and independent check |
| 150–180 min | Discuss results (20), exit ticket (5), feedback (3), closing (2) |

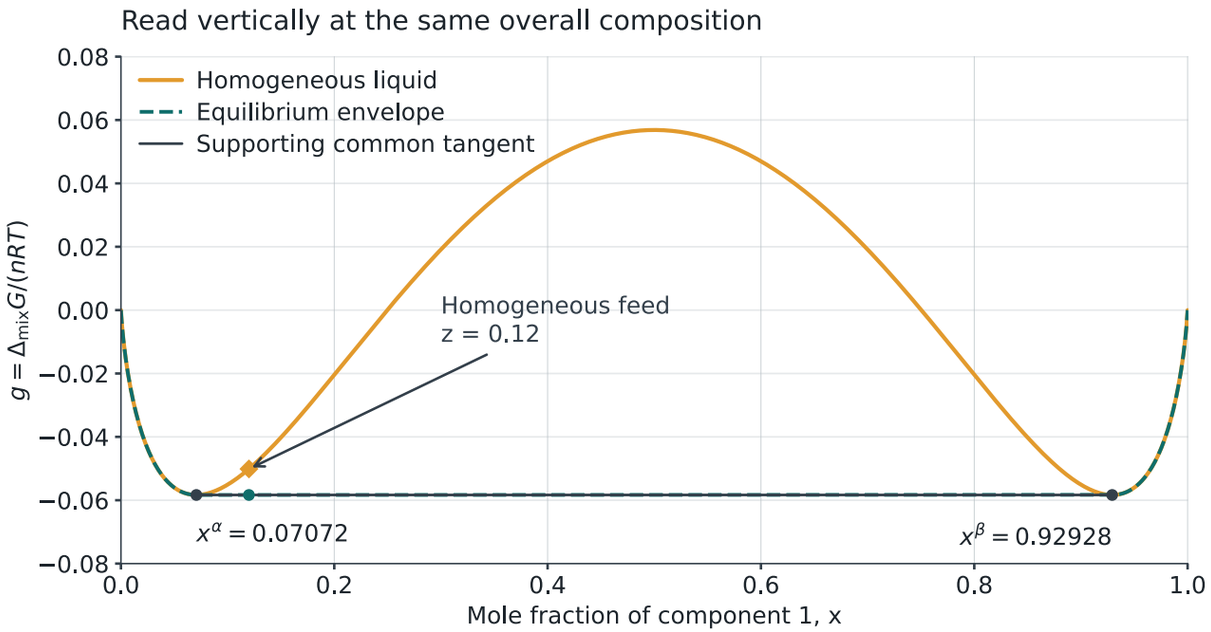
# Week 2 · What did the VLE measurement tell us?

A VLE measurement gives $P = 100$ kPa, $x_1 = 0.40$, $y_1 = 0.60$. Assume an ideal vapor, negligible Poynting correction and a pure-liquid reference state.

$$y_1 P = x_1 \gamma_1 P_1^{\mathrm{sat}}, \qquad a_1 = x_1 \gamma_1 .$$

1. Use $P_1^{\mathrm{sat}} = 120$ kPa. Find $\gamma_1$ and $a_1$.
2. A second source gives 150 kPa at the same temperature. Predict which way your inferred $\gamma_1$ moves.
3. Before fitting the data, which Antoine conventions would you check?

# Week 3 · Would this feed split into two liquids?




Mixing Gibbs energy and the supporting common tangent for the Week 3 Margules liquid at A=3.

At $z_1 = 0.12$, compare the curve and the supporting line at the **same overall composition**. Which state has lower Gibbs energy?

The homogeneous curve has positive local curvature here. What does the tangent tell you that curvature alone cannot?

A Gibbs–Duhem area test examines the VLE data. How could errors of opposite sign affect its signed integral?

# Week 4 · Can the vessel supply that much gas?

At 300 K, a 1.00 L vessel contains 0.00200 mol of volatile A. Its equilibrium sublimation pressure is 10.0 kPa. Use $R = 8.314$ kPa L mol$^{-1}$ K$^{-1}$ and assume an ideal gas.

1. Calculate the gas inventory required for $P_A = 10.0$ kPa. Will any bulk solid remain?
2. Calculate the actual $P_A$. What happens to $P_A$ if you add an inert gas at fixed $T, V$?
3. In a separate adsorption vessel, an adsorbent of mass $m_s$ holds $q(P_A, T)$ mol per kg. Which term must you add to the gas inventory to close the balance?

# Week 5 · Change pressure, then change temperature

Lab 12 used 1 mol of A₂ initially, zero A, no inert, and $p^\circ = 100$ kPa. Use the same ideal-gas reaction in all three cases.

| Case | $T$ (K) | $P$ (kPa) | $K$ | Equilibrium extent (mol) |
|---|---|---|---|---|
| 12A | 350 | 100 | 3.953 | 0.7050 |
| 12B | 350 | 400 | 3.953 | 0.4451 |
| 12C | 400 | 100 | 33.861 | 0.9457 |

Which pair isolates pressure? Which isolates temperature? For 12C assume a constant positive reaction enthalpy of 50 kJ mol$^{-1}$. Explain the change in $K$.

With a partner, use the extent in 12B to check $2n_{\mathrm{A_2}} + n_{\mathrm{A}}$ against the feed.

# Check your four predictions

| Week | Compare with your calculation or prediction |
|---|---|
| 2 · VLE | With $P_1^{\mathrm{sat}} = 120$ kPa, $\gamma_1 = 1.25$ and $a_1 = 0.50$. With $P_1^{\mathrm{sat}} = 150$ kPa they become 1.00 and 0.40. Check Antoine units, log base and temperature range. |
| 3 · Stability | At $z_1 = 0.12$ the two-liquid line lies lower. Positive local curvature describes small fluctuations. Opposite errors can cancel in the signed area test. |
| 4 · Inventory | Saturation requires 0.00401 mol. All 0.00200 mol enters the gas, giving $P_A = 4.99$ kPa. Adding ideal inert at fixed $T, V$ raises total pressure while $P_A$ stays at 4.99 kPa. For adsorption add $m_s q$ mol. |
| 5 · Reaction | 12A–12B isolate pressure. 12A–12C isolate temperature. Heating raises $K$ for this endothermic reaction. In 12B, $2(0.5549) + 0.8902 = 2.0000$ mol of A atoms. |

# The question for today

What determines the sign of a concentration-cell potential?

Week 5 gave us $\Delta_r G = RT\ln(Q/K)$. A reversible cell turns that reaction driving force into a potential difference, $E = -\Delta_r G/(nF)$.

**Predict on paper.** For the reaction as written, what sign should $E$ have when $Q < K$? What voltage would you measure at complete cell equilibrium, $Q = K$?

Pre-class reading and preparation questions

# Two reduction half-reactions

Enter both half-reactions as reductions. Reverse the left reaction and add the right after matching electron counts.

$$E^\circ_{cell} = E^\circ_{right} - E^\circ_{left}$$

Both potentials need the same reference electrode and temperature. Multiplying a half-reaction does not multiply its potential.

# Nernst equation from reaction thermodynamics

Combine ΔrG=ΔrG°+RTlnQ with ΔrG=−nFE:

$$E = E^\circ - \frac{RT}{nF} \ln Q$$

At 298.15 K,

$$E = E^\circ - \frac{0.05915935\ \mathrm{V}}{n} \log_{10} Q$$

Changing T requires valid E° data at the new T.

# A concentration-cell calculation

Use the same $\mathrm{M}^{2+}/\mathrm{M}$ couple on both sides at 298.15 K. For oxidation on the left and reduction on the right, $Q = a_{\mathrm{L}}/a_{\mathrm{R}}$. Take $a_{\mathrm{L}} = 0.01$, $a_{\mathrm{R}} = 1$, $n = 2$, and neglect the junction potential.

$$Q = 0.01, \qquad E = +0.05915935\ \mathrm{V}$$

E°=0, K=1, but E is nonzero. Reversing the concentrations changes the sign of E. Equal activities give E=0.

ΔrG≈−11.416 kJ per mol of the written reaction.

# An independent check

Use dimensionless activities. Concentration is an approximation with a stated standard concentration and activity coefficient.

Use Eright−Eleft and deltaG=−nFE with the balanced electron count. Equal activities must give zero potential for the same reduction couple.

# One demonstration · 10 minutes

## Open the core lab

Open the concentration-cell preset on the projector. On paper, make three rows for the original, exchanged and equal activities. If your device is offline, use the projected inputs and complete the same hand check. Predict the sign, exchange left and right activities, then set them equal. Keep T and reference conventions fixed.

Keep one saved baseline for the pair investigation.

# Pair investigation · 70 minutes

- 10 min: predict and record the baseline.
- 25 min: change one input and calculate.
- 20 min: exchange roles and check the result by hand.
- 15 min: prepare one plot or table and a short explanation.

In Lab 13, compare a concentration cell with the activities exchanged. Predict and explain the sign of E, then independently check deltaG=−nFE.

# Discussion · 20 minutes

What determines the sign of a concentration-cell potential?

Compare results and discuss unexpected behavior. Use one numerical check to support your explanation.

# After class · 30–40 minutes

In Lab 13, compare a concentration cell with the activities exchanged. Predict and explain the sign of E, then independently check deltaG=−nFE.

Assessment 6 and independent checkpoint

Retain one study JSON, one comparison and a short explanation.

Optional: Session appendix.

Learning path · Offline package

# Exit ticket + feedback · 8 minutes

**5 min · Explain and check**

Answer the session question in the form. Give one result or independent check from today’s lab, with units.

**3 min · Help improve the next class**

Rate the pace, report lab access, and suggest one improvement.

Thai or English. No name, ID or email requested.

Wait for **“Response received.”** We will use the final 2 minutes to close the class.

Scan to submit Session 6 exit ticket and feedback.

Open Session 6 form